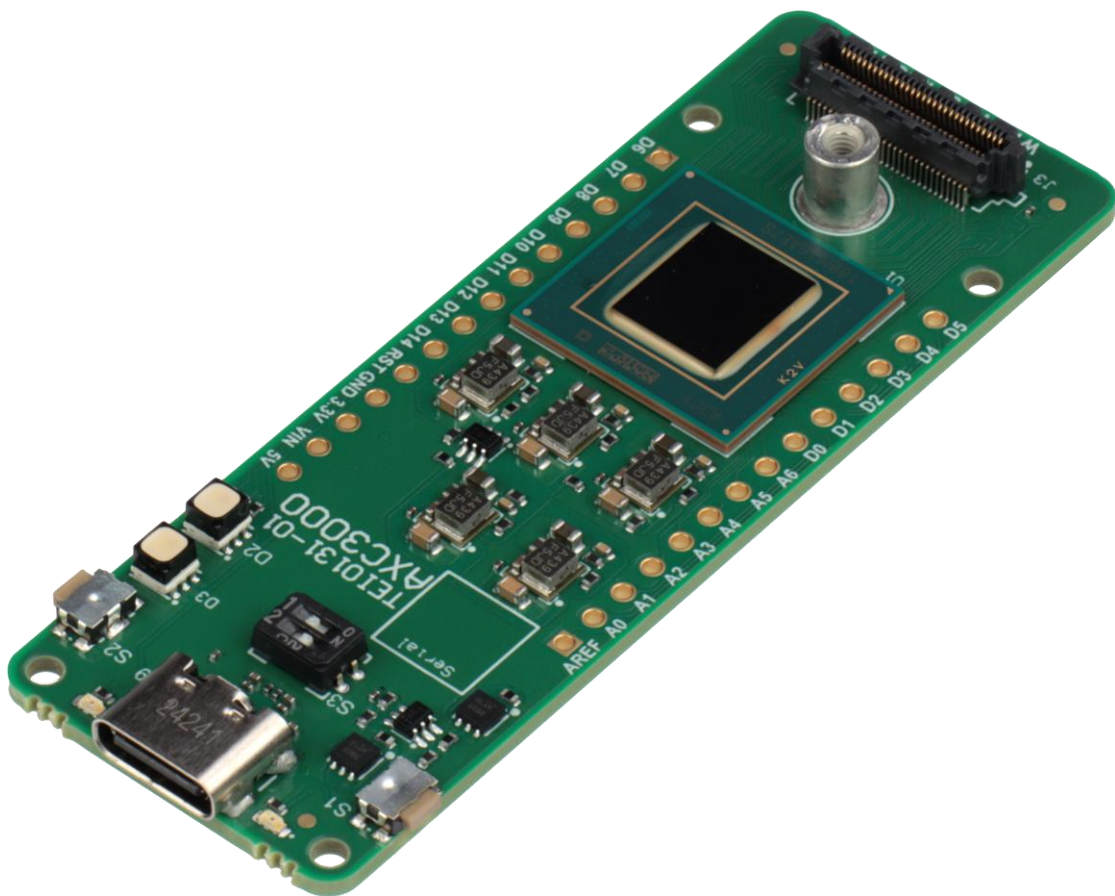




Installation Guide (North America)



Document Control

Document Version:	Version 1.0
Document Date:	10/27/2025
Document Author(s):	Steven Kravatsky, Erika Peter, Naji Naufel,, ESC Team
Document Classification:	Released
Document Distribution:	This document is still under development. All specifications, procedures, and processes described in this document are subject to change without prior notice
Prior Version History:	

Please read the legal disclaimer at the end of this document.

Contents

1	Workshop scenarios	4
2	CloudLab scenario	5
2.1	Account registration	5
2.2	Hardware requirements	5
2.3	Software requirements	5
3	Local PC (verification only) scenario	6
3.1	Account registration	6
3.2	Hardware requirements	6
3.3	Software requirements	6
4	Local PC scenario	12
4.1	Account registration	12
4.2	Hardware requirements	12
4.3	Software requirements	12
5	Licensing	28
5.1	Quartus Prime Pro Edition Evaluation Licenses	28
6	Legal Disclaimer	35

1 Workshop scenarios

There are three different scenarios in which the Agilex 3 labs can be implemented and verified. They are:

CloudLabs – build in the cloud and verify with board farm hardware

Local PC (verification only) – build in the cloud, verify on local hardware

Local PC – build with a local PC and verify with local hardware

All three have different installation requirements. Each scenario's requirements are described in this document. Please choose only one scenario and implement its requirements.

Note: The North America Workshop events will only support the CloudLabs and Local PC (verification only) scenarios on the workshop day. The Local PC scenario is documented so the user may continue their learning after the workshop day is complete.

For a local PC installation, the files are large, so the internet speed needs to be fast.

2 CloudLab scenario

This scenario does not require any installation.

2.1 Account registration

ONE WARE AI: The lab user guide provides details on obtaining an account.

FPGA AI Suite requires access to Google Colab. This is where the AI model will be constructed and trained with a dataset. Follow the steps below to obtain a Google account. You may use an existing account, if available.

Click on [this link](#) to register a new Google account. Click Create account.

2.2 Hardware requirements

- Personal computer or laptop running Windows 11 with at least an Intel i5 core (or equivalent), 8GB RAM.

2.3 Software requirements

- Web browser.

3 Local PC (verification only) scenario

Two AI labs do require obtaining account credentials for online tool sets. Only follow the instructions below if you have chosen to do that lab.

3.1 Account registration

ONE WARE AI: The lab user guide provides details on obtaining an account.

FPGA AI Suite requires access to Google Colab. This is where the AI model will be constructed and trained with a dataset. Follow the steps below to obtain a Google account. You may use an existing account, if available.

Click on [this link](#) to register a new Google account. Click Create account.

3.2 Hardware requirements

- Personal computer or laptop running Windows 10/11 with at least an Intel i5 core (or equivalent), 8GB RAM, 10GB hard drive space.
- AXC3000 Board
- USB C cable

3.3 Software requirements

- Web browser
- Quartus Prime 25.1.1 Pro Standalone Programmer.
- Terminal emulator (Putty or Tera term)
- Python 3.13

3.3.1 Install Quartus Prime Pro Edition Programmer and Tools (25.1.1). Navigate to the Quartus Prime Pro 25.1.1 install page. [Click on this link](#).

3.3.1.1 Scroll down and select the Individual Files tab under the Downloads section.

Downloads

[Installer \(Recommended\)](#)
[Individual Files](#)
[Copyleft Licensed Source](#)

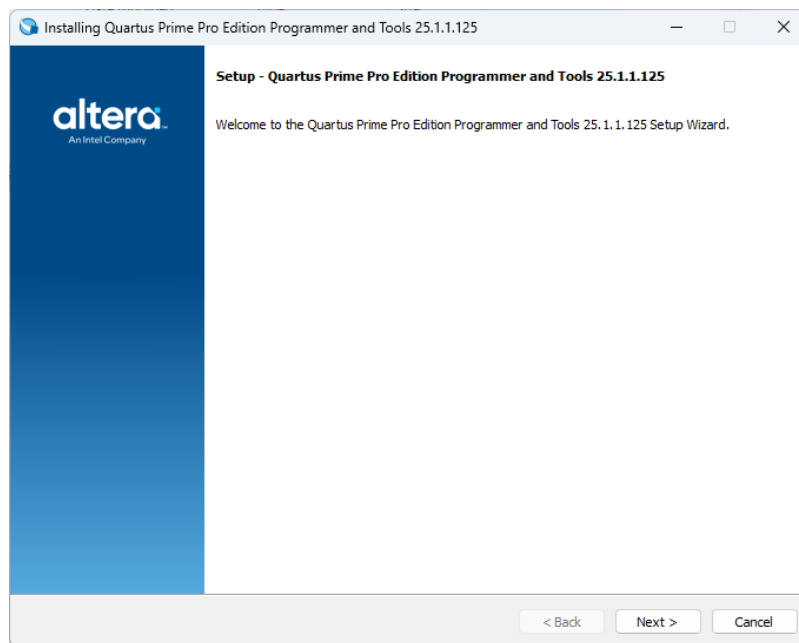
3.3.1.2 Scroll further to the Add-On and Stand-Alone Software section. Click on **Quartus Prime Pro Edition Programmer and Tools** to begin the download.

Quartus® Prime Pro Edition Programmer and Tools	
Download QuartusProProgrammerSetup-25.1.1.125-windows.exe	Size: 1.5 GB SHA1: c12c95bc01e484e85555747b834ffa97ae39b201

3.3.1.3 Click on the downloaded file to begin the installation. The file will typically be located in the Downloads folder by default.

 QuartusProProgrammerSetup-25.1.1.125-windows	8/25/2025 12:00 PM	Application	1,571,762 KB
--	--------------------	-------------	--------------

3.3.1.4 Follow the installation instructions.



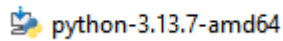
3.3.2 Install a terminal emulator.

- Install [TeraTerm](#) or
- Install [Putty](#)

3.3.3 Install Python

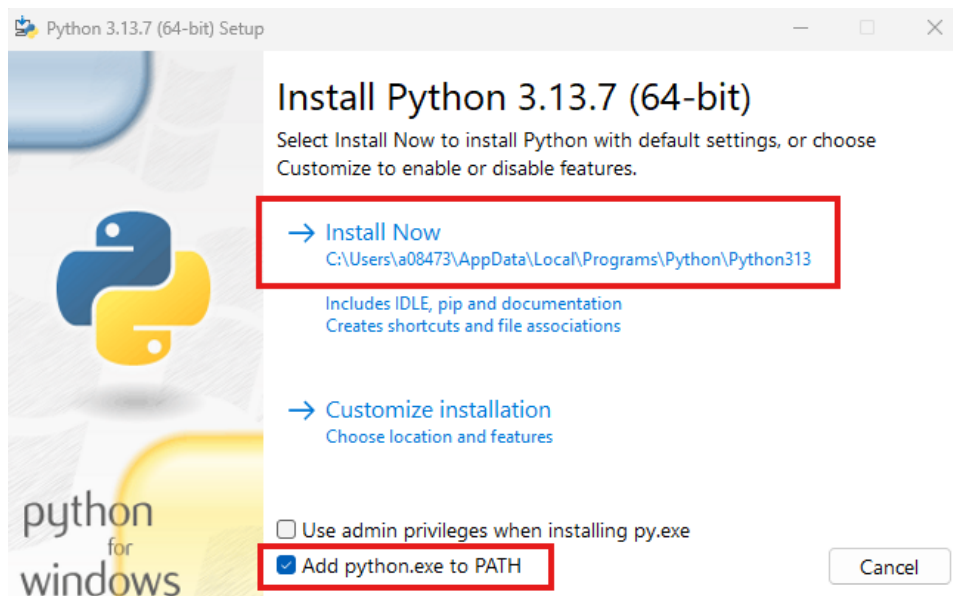
3.3.3.1 Download the [installer](#)

3.3.3.2 Right click the **installer** and select open.



3.3.3.3 Check the box ' **Add python.exe to PATH**'.

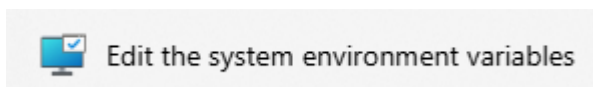
3.3.3.4 Select **Install Now**



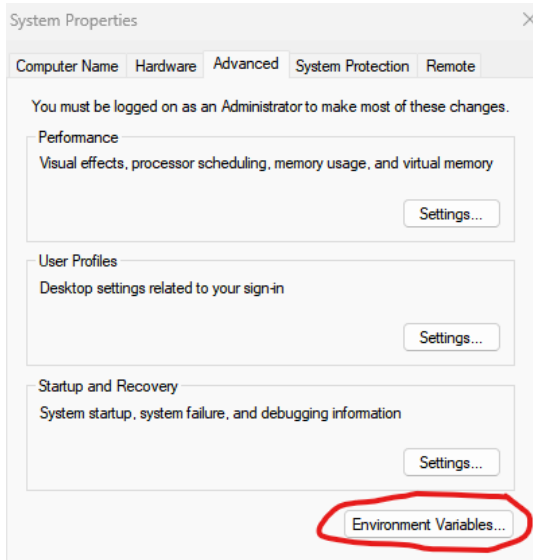
3.3.4 Add Environment Variable for System Console

3.3.4.1 Type Environment in the Windows search bar.

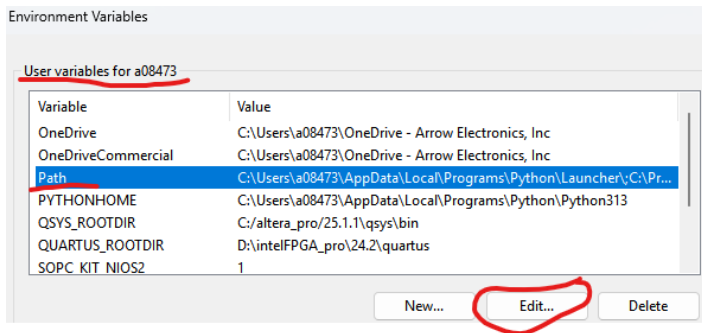
3.3.4.2 Select Edit **the system environment variables**



3.3.4.3 Press the Environment Variables button



3.3.4.4 Select PATH variable. Press Edit.



3.3.4.5 Press New

3.3.4.6 Add the following PATH

c:\altera_pro\25.1.1\qprogrammer\syscon\bin

3.3.5 Install Python Installer pip

3.3.5.1 Open a CMD shell and execute the following commands. If the 'python' command is not recognized, replace it with 'py'.

```
curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py
python get-pip.py
```

3.3.5.2 Install Python libraries. Execute the following in the CMD shell.

```
pip install numpy
pip install willow
pip install matplotlib
pip install telnetlib3
pip install serial
pip install pyserial
```

3.3.6 Install git for Windows

3.3.6.1 Use the Windows search bar to find **PowerShell**. Open PowerShell.

3.3.6.2 Copy the following **line of text** into the PowerShell and press enter.

```
winget install --id Git.Git -e --source winget
```

3.3.7 Installing USB-Blaster III on native Linux machines.

The Altera tools in general support Linux, but given the variety of distributions you may experience some challenges with programming cable(USB Blaster) drivers.

The following description has been verified on:

- Ubuntu 24.03.3 fresh install with default settings
- Quartus 25.3 fresh install into default location under home directory

```
# Step 1: Unplug USB Blaster cable and run these commands:
sudo tee /etc/udev/rules.d/92-usbblaster.rules >/dev/null <<'EOF'
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6001", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6002", MODE="0666"
```

```
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6003", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6010", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6810", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6020", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6022", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6024", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6025", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6026", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602C", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602D", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602E", MODE="0666"
EOF
```

```
sudo udevadm control --reload-rules
sudo udevadm trigger
```

```
# Step 2: Replug the USB Blaster cable
```

```
# Step 3 (Optional)
```

```
# Make sure <altera_install_dir>/quartus/bin is in the PATH to find jtagd and
jtagconfig executables...
```

```
pkill jtagd
```

```
jtagd --user-start
```

```
jtagconfig
```

4 Local PC scenario

Two AI labs do require obtaining account credentials for online tool sets. Only follow the instructions below if you have chosen to do that lab.

4.1 Account registration

ONE WARE AI: The lab user guide provides details on obtaining an account.

FPGA AI Suite requires access to Google Colab. This is where the AI model will be constructed and trained with a dataset. Follow the steps below to obtain a Google account. You may use an existing account, if available.

Click on [this link](#) to register a new Google account. Click Create account.

4.2 Hardware requirements

- Personal computer or laptop running Windows 10/11 with at least an Intel i7 core (or equivalent), 32GB RAM, 250GB hard drive space or more.
- AXC3000 Board
- USB C cable

4.3 Software requirements

- Quartus Prime 25.1.1 Pro.
- WSL2 Ubuntu 22.04 virtual machine
- FPGA AI Suite
- ONE WARE Studio
- Terminal emulator (Putty or Tera term)

4.3.1 Quartus Prime 25.1.1 Pro.

4.3.1.1 Download Quartus Prime Pro 25.1.1 at the [following link](#).

Date	Software Type	Software Package	Version	Operating Systems
8/11/2025	FPGA Development Tools	Quartus® Prime Pro	25.1.1 (Latest)	Windows

4.3.1.2 **Select the Individual Files** to reduce the amount of space needed on a computer. (The complete install could be downloaded, if wished but it will use more space.)

Downloads

[Installer \(Recommended\)](#)
[Individual Files](#)
[Copyleft Licensed Source](#)

4.3.1.3 Select and download the Quartus Prime Pro Edition Part 1

Quartus® Prime Pro Edition Part 1

Download
QuartusProSetup-25.1.1.125-windows.exe

4.3.1.4 When the Legal Disclaimer appears, read it and select **Accept**


Software License Agreement

Legal Disclaimer

PLEASE NOTE: This version of software ("Software") does not contain the latest functional and security updates. In order to use this version, you must first acknowledge the following term, which supplements and supersedes any inconsistent provision in the version of the Intel® FPGA Software License Subscription Agreement for the product (e.g., Intel® Quartus® Prime Software, Intel® HLS Compiler, Intel® FPGA SDK for OpenCL™, DSP Builder for Intel® FPGAs, or Advanced Link Analyzer) with which you use the Software:

Intel does not give or enter into any condition, warranty, or other term:

- i. with respect to any malfunctions or other errors in its Software caused by virus, infection, worm or similar malicious code not developed or introduced by Intel; or
- ii. to the effect that any Software will protect against all possible security threats, including intentional misconduct by third parties.

Intel is not liable for any downtime or service interruption, for any lost or stolen data or systems, or for any other damages arising out of or relating to any such actions or intrusions or resulting from use of Software. Intel does not give or enter into any condition, warranty, or other term with respect to interoperability.

Intel does not warrant or assume responsibility for the accuracy or completeness of any information, text, graphics, links or other items within the Software. Please click "Accept" below to continue the download process.

Accept

Reject

[Terms of Use](#) |
 [Privacy](#) |
 [Legal Information](#) |
 © Intel Corporation

4.3.1.5 It should start downloading.

4.3.1.6 **Select** the back button in the Browser to go back to the other files

4.3.1.7 Select the following files for download under the Individual files tab in the Individual files section.

Quartus® Prime Pro Edition Part 1

Download
QuartusProSetup-25.1.1.125-windows.exe

Quartus® Prime Pro Edition Part 2

Download
QuartusProSetup-part2-25.1.1.125-windows.qdz

Questa*-Intel FPGA and Starter Editions

Download
QuestaSetup-25.1.1.125-windows.exe

4.3.1.8 Select the following files for download under the Individual files tab in the Add-On and Stand-Alone Software files section.

Ashling* RiscFree* IDE for Altera

Download
RiscFreeSetup-25.1.1.125-windows.exe

Power Thermal Calculator

Download
ptc-25.1.1.125-windows.exe

4.3.1.9 Select the following files for download under the Individual files tab in the Devices files section.

Agilex™ 3 device support (Requires Agilex™ common files)

Download
agilex3-25.1.1.125.qdz

Agilex™ common files

Download
agilex_common-25.1.1.125.qdz

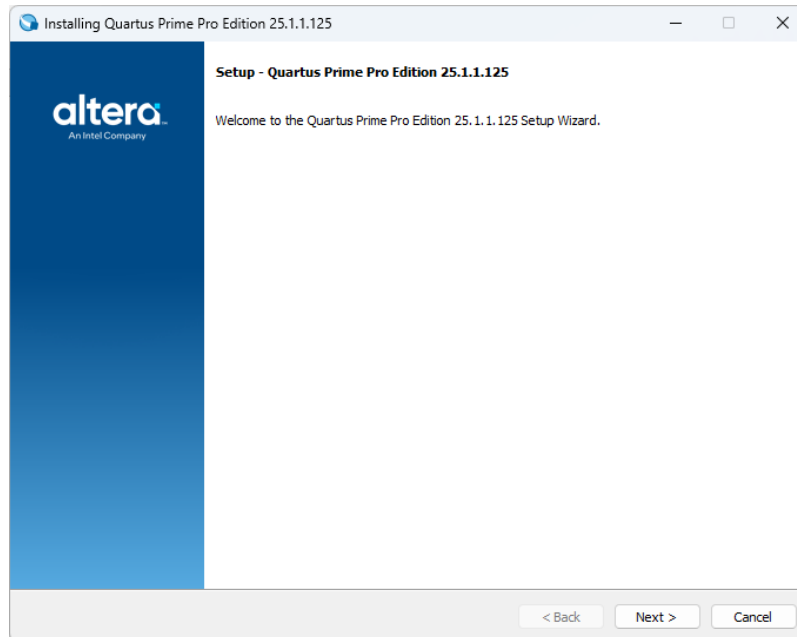
4.3.1.10 The downloaded files when viewed in Windows Explorer.

	QuartusProSetup-25.1.1.125-windows	8/25/2025 6:40 PM	Application
	QuartusProSetup-part2-25.1.1.125-win...	8/25/2025 6:36 PM	QDZ File
	agilex_common-25.1.1.125.qdz	8/25/2025 6:35 PM	QDZ File
	RiscFreeSetup-25.1.1.125-windows	8/25/2025 6:34 PM	Application
	QuestaSetup-25.1.1.125-windows	8/25/2025 6:34 PM	Application
	ptc-25.1.1.125-windows	8/25/2025 6:32 PM	Application
	agilex3-25.1.1.125.qdz	8/25/2025 6:29 PM	QDZ File

4.3.1.11 Select QuartusProSetup-25.1.1.125-windows. Right click, select open.

4.3.1.12 On some computers, it may be possible that you need to sign in for admin access in order to install software. It may be a while for the Quartus to appear.

4.3.1.13 The Quartus setup window will appear.



4.3.1.14 Select **Next**.

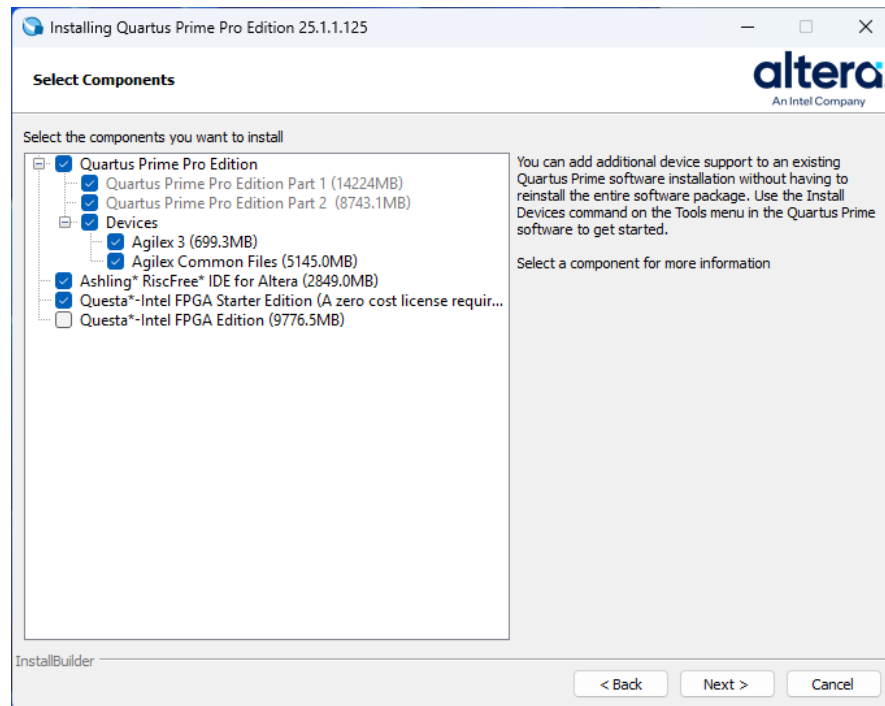
4.3.1.15 Read the License Agreement and select **I accept the agreement**.

4.3.1.16 Select **Next**.

4.3.1.17 Select the **default** directory.

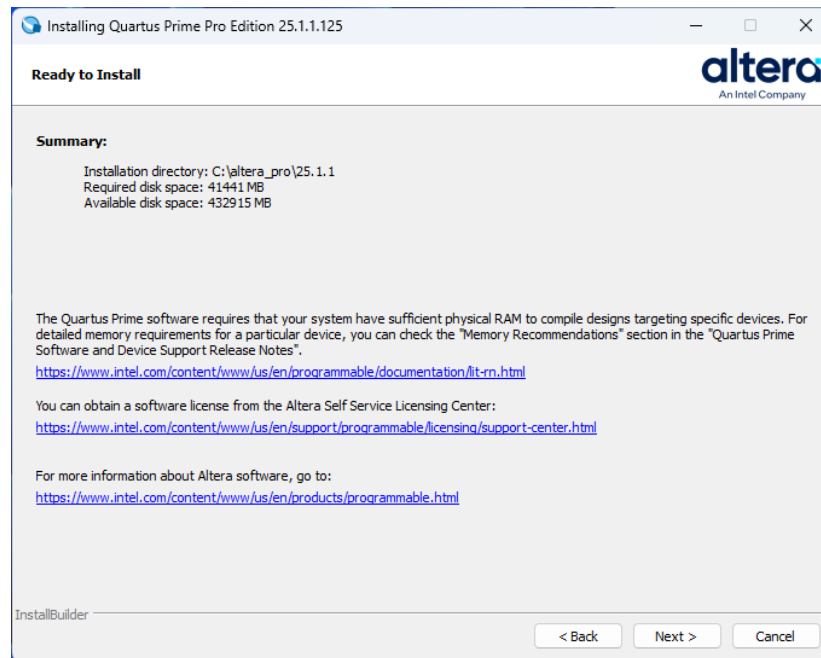
4.3.1.18 Select **Next**

4.3.1.19 Ensure that the following items are enabled

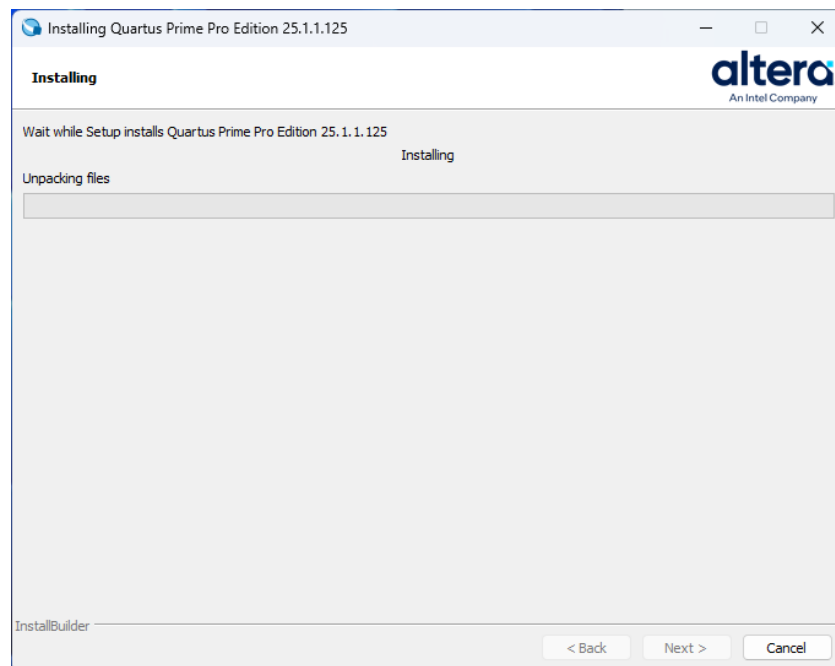


4.3.1.20 Select **Next**

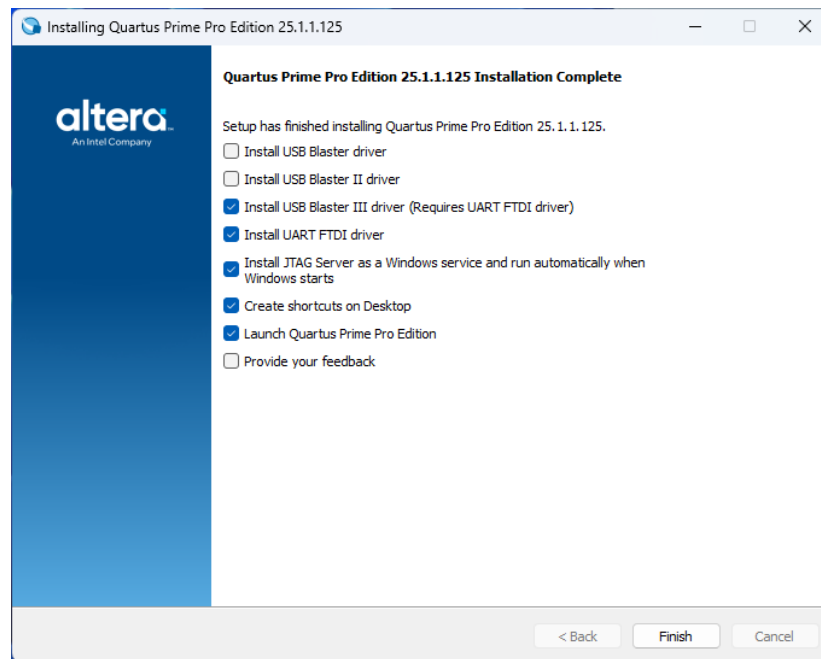
4.3.1.21 The Summary window will appear.



4.3.1.22 The install can take a while.



4.3.1.23 The setup installer will indicate when it is complete. A driver installation window will be shown. Uncheck the first two boxes.



4.3.1.24 Select Finish. Click through the driver install windows until complete.

4.3.2 Install WSL2 Ubuntu 22-04

4.3.2.1 Open Windows PowerShell with administrator privileges. Execute the following instructions to install Ubuntu 22.04 LTS

- `dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart`
- `dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart`
- `wsl --set-default-version 2`
- reboot windows and open PowerShell with admin. priviledges
- `wsl --install -d Ubuntu-22.04`
 - select username = **altera**, password = **agilex3**, when prompted

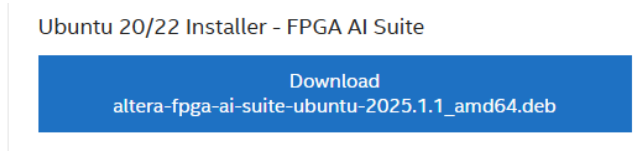
4.3.2.2 Invoke WSL2 with Ubuntu 22.04 LTS. Type Ubuntu in the Windows search bar. Click on the Ubuntu 22.04 LTS icon. Execute the following instructions in the Ubuntu shell.

- `sudo apt update`
- `sudo apt install wget`
- `wget https://apt.kitware.com/ubuntu/pool/main/c/cmake/cmake\_3.31.6-0kitware1ubuntu22.04.1\_amd64.deb`
- `wget https://apt.kitware.com/ubuntu/pool/main/c/cmake/cmake-data\_3.31.6-0kitware1ubuntu22.04.1\_all.deb`

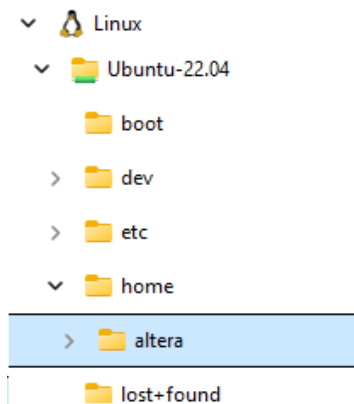
- `sudo dpkg -i cmake-data_3.31.6-0kitware1ubuntu22.04.1_all.deb`
- `sudo dpkg -i cmake_3.31.6-0kitware1ubuntu22.04.1_amd64.deb`

4.3.3 Install Altera FPGA AI Suite

4.3.3.1 Download Altera FPGA AI Suite. Click on this [link](#) to navigate to the download page. Click on Ubuntu 20/22 Installer - FPGA AI Suite.



4.3.3.2 Drop and drag the installation file from Windows into the Ubuntu filesystem. Open explorer in windows and navigate to the Linux icon. Expand and select the /home/altera folder. Drop the installation file into the /home/altera folder.



4.3.3.3 Copy the installation file to /opt/altera. If prompted, the password is **agilex3**. Execute the following instructions in the Ubuntu shell.

- `sudo mkdir opt/altera`
- `cd /opt/altera`
- `sudo cp ~/altera-fpga-ai-suite-ubuntu-2025.1.1_amd64.deb /opt/altera`

4.3.3.4 Install FPGA AI Suite. Execute the following instructions in the Ubuntu shell.

- `sudo apt install --reinstall libappstream4`
- `sudo apt install /opt/altera/altera-fpga-ai-suite-ubuntu-2025.1.1_amd64.deb`
- if a dependency **error or warning** occurs do the following, `sudo apt --fix-broken install`

4.3.4 Install the OpenVINO toolkit

4.3.4.1 Install required OpenVINO components. Upgrade to Python 3.12. Execute the following instructions.

- `sudo apt update && sudo apt upgrade -y`

- `sudo add-apt-repository ppa:deadsnakes/ppa -y`
- `sudo apt update`
- `sudo apt install python3.12 python3.12-venv -y`
- `python3.12 -V`

4.3.4.2 Create a new Python virtual environment. Execute the following instructions.

- `mkdir ~/build-openvino-dev && cd ~/build-openvino-dev`
- `python3.12 -m venv openvino_env`
- `source openvino_env/bin/activate`
-

4.3.4.3 Install the OpenVino 2024.6 Runtime. Execute the following instructions.

- `sudo mkdir /opt/intel`
- `mkdir ~/Downloads && cd ~/Downloads`

4.3.4.4 Download the OpenVINO 2024.6 Runtime archive file. Execute the following instructions.

```
curl -L https://storage.openvinotoolkit.org/\
repositories/openvino/packages/2024.6/linux/\
l_openvino_toolkit_ubuntu20_2024.6.0.17404.4c0f47d2335_x86_64.tgz \
--output openvino_2024.6.0.tgz
```

- `tar -xf openvino_2024.6.0.tgz`
- `sudo mv l_openvino_toolkit_ubuntu20_2024.6.0.17404.4c0f47d2335_x86_64 /opt/intel/openvino_2024.6.0`

4.3.4.5 Install system dependencies required by OpenVINO runtime.

- `cd /opt/intel/openvino_2024.6.0/install_dependencies/`
- `sudo -E ./install_openvino_dependencies.sh`

4.3.4.6 Create a symbolic link.

- `cd /opt/intel`
- `sudo ln -s openvino_2024.6.0 openvino_2024`

4.3.4.7 Configure openvino runtime environment variables.

- `source /opt/intel/openvino_2024/setupvars.sh`

4.3.4.8 Install openvino development tools.

- `python3 -m pip install --upgrade pip`
- `pip install "openvino-dev[caffe,pytorch,tensorflow]==2024.6.0"`

4.3.4.9 Install OpenCV.

- `sudo apt update`
- `sudo apt install libopencv-dev python3-opencv`

- `pip install opencv-python` (ignore numpy error??)

4.3.4.10 Configure OpenVINO environment.

- `source ~/build-openvino-dev/openvino_env/bin/activate`
- `unset python_version`
- `source /opt/intel/openvino_2024/setupvars.sh`

4.3.4.11 Add pytorch libraries.

- `sudo apt update`
- `sudo apt install python3-pip -y`
- `pip3 install torch onnx torchsummary`

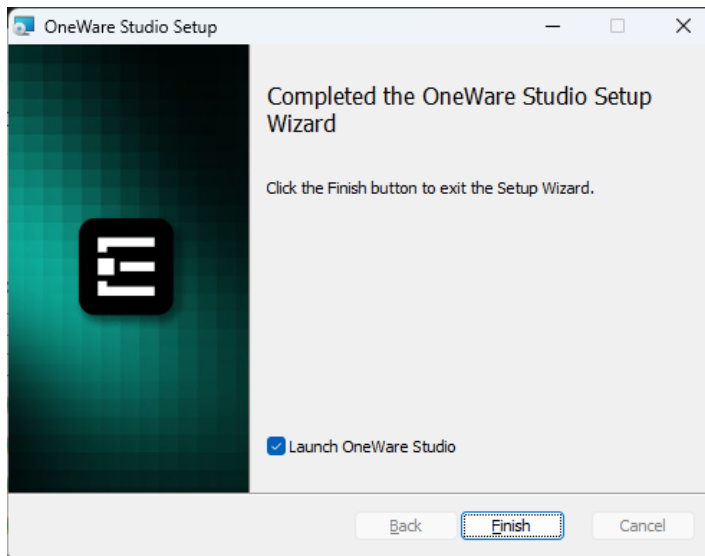
4.3.5 ONE WARE Studio

Only install if you will be completing the ONE WARE AI lab.

4.3.5.1 Click on [this link](#) to download the installer.

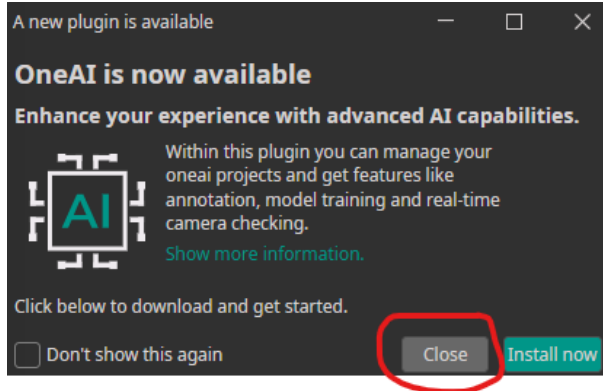
4.3.5.2 Then right click the file and select open.

4.3.5.3 Accept the license agreement and click through the installer. Click Finish.

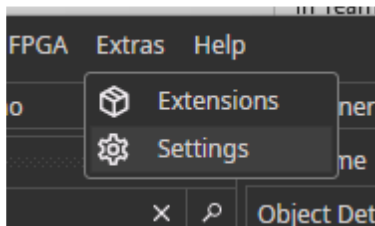


4.3.5.4 ONE WARE Studio will automatically launch. Complete the setup with the next steps.

4.3.5.5 Select **Close** when the OneAI plugin is recommended.



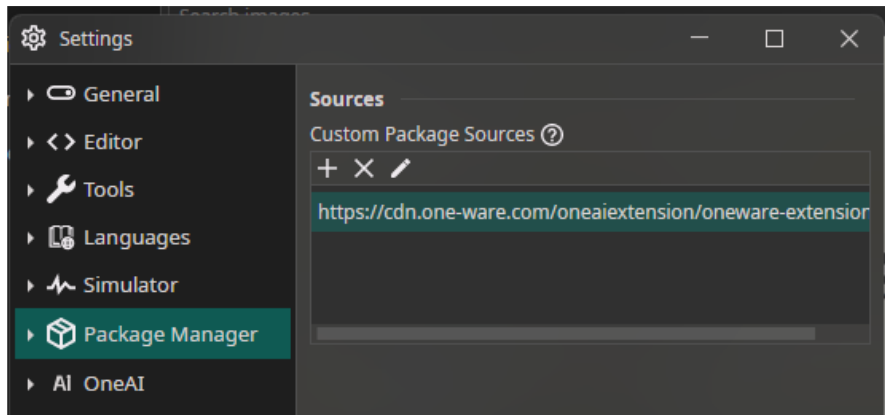
4.3.5.6 Select the Extras → Settings menu.



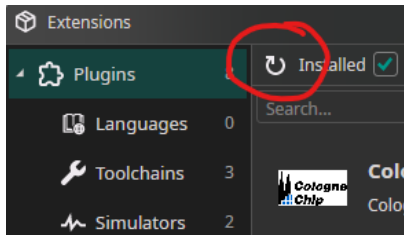
4.3.5.7 Go to the Package Manager tab. Add this path

- <https://cdn.one-ware.com/oneaiextension/oneware-extension.json>

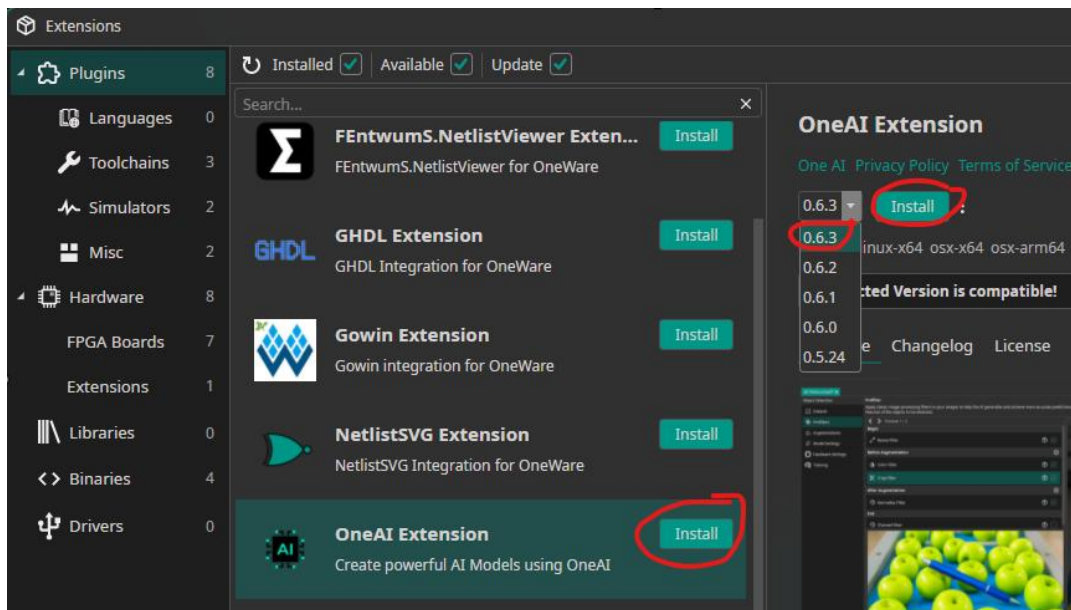
Click the Save button.



4.3.5.8 Select the Extras → Extensions menu. Click on the refresh button.



4.3.5.9 Install The ONE AI Extension. Click Install. Select version 0.63 from the pulldown menu. Click Install.



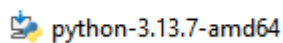
4.3.6 Install a terminal emulator.

- Install [TeraTerm](#) or
- Install [Putty](#)

4.3.7 Install Python

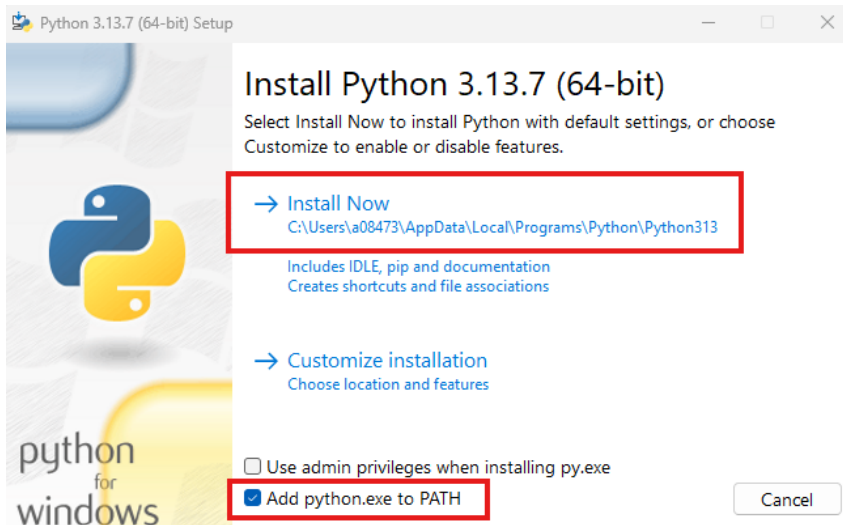
4.3.7.1 Download the [installer](#)

4.3.7.2 Right click the **installer** and select open.



4.3.7.3 Check the box 'Add python.exe to PATH'.

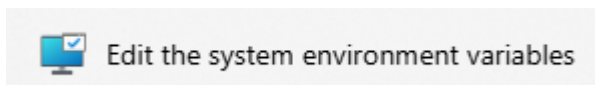
4.3.7.4 Select **Install Now**



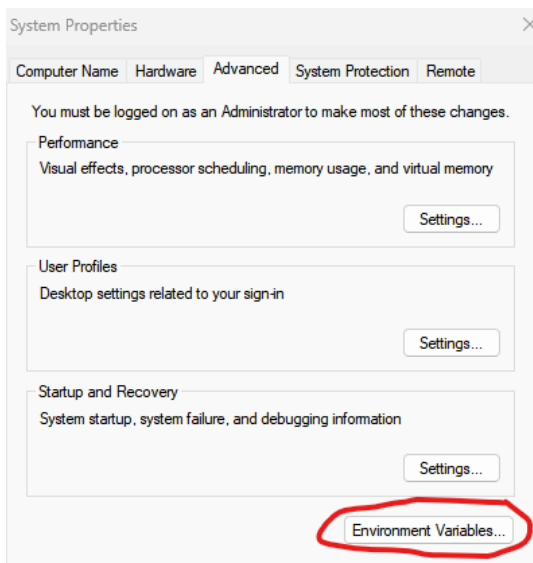
4.3.8 Add Environment Variable for System Console

4.3.8.1 Type Environment in the Windows search bar.

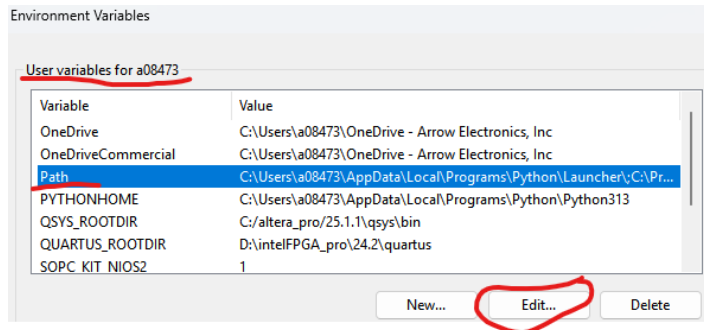
4.3.8.2 Select Edit **the system environment variables**



4.3.8.3 Press the Environment Variables button



4.3.8.4 Select PATH variable. Press Edit.



4.3.8.5 Press New

4.3.8.6 Add the following PATH

```
c:\altera_pro\25.1.1\syscon\bin
```

4.3.9 Install Python Installer pip

4.3.9.1 Open a CMD shell and execute the following commands. If the 'python' command is not recognized, replace it with 'py'.

```
curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py
python get-pip.py
```

4.3.9.2 Install Python libraries. Execute the following in the CMD shell.

```
pip install numpy
pip install willow
pip install matplotlib
pip install telnetlib3
pip install serial
pip install pyserial
```

4.3.10 Install git for Windows

4.3.10.1 Use the Windows search bar to find **PowerShell**. Open PowerShell.

4.3.10.2 Copy the following **line of text** into the PowerShell and press enter.

```
winget install --id Git.Git -e --source winget
```

4.3.11 Installing USB-Blaster III on native Linux machines.

The Altera tools in general support Linux, but given the variety of distributions you may experience some challenges with programming cable(USB Blaster) drivers.

The following description has been verified on:

- Ubuntu 24.03.3 fresh install with default settings
- Quartus 25.3 fresh install into default location under home directory

```
# Step 1: Unplug USB Blaster cable and run these commands:
sudo tee /etc/udev/rules.d/92-usbblaster.rules >/dev/null <<'EOF'
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6001", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6002", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6003", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6010", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6810", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6020", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6022", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6024", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6025", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6026", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602C", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602D", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602E", MODE="0666"
EOF

sudo udevadm control --reload-rules
sudo udevadm trigger

# Step 2: Replug the USB Blaster cable

# Step 3 (Optional)
#   Make sure <altera_install_dir>/quartus/bin is in the PATH to find jtagd and
#   jtagconfig executables...
pkill jtagd
jtagd --user-start
jtagconfig
```

5 Licensing

Quartus Prime Pro requires a free license for Agilex 3 devices. Licenses are also required for the Nios V/g and MIPI CSI-2 IP cores. The Questa are requires a license.

5.1 Identify the PC NIC ID

5.1.1 Open a Windows PowerShell

5.1.2 Type ipconfig /all. The NIC is the Physical Address of the adapter. This will be used to create a new computer for licensing purposes.

```
Ethernet adapter Ethernet:

Connection-specific DNS Suffix  . : 
Description . . . . . : Intel(R) Ethernet Connection (22) I219-LM
Physical Address. . . . . : AC-B4-80-94-58-7B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

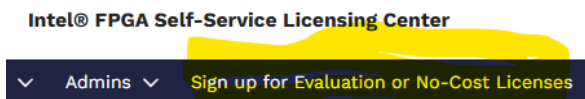
5.2 Create Evaluation Licenses

5.2.1 Navigate to the [Self Service Licensing Center](#)

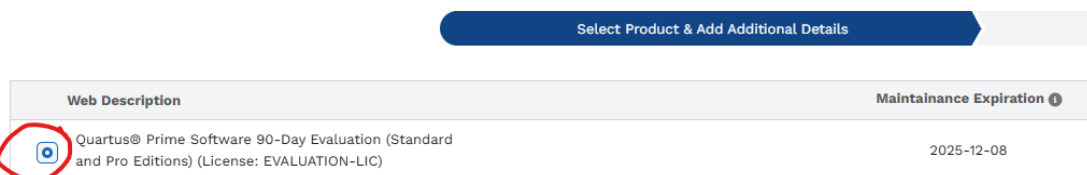
5.2.2 Sign in to an existing account or Enroll to get a new account.



5.2.3 Sign up for Evaluation or No-Cost Licenses



5.2.4 Select Quartus Prime Software 90-Day Evaluation License



5.2.4.1 Scroll to the bottom of the page

5.2.4.2 Select Intel FPGA AI Suite and Questa-Intel FPGA Edition

Select Optional Products

☐ Select/Unselect All Options

☐ Intel® Cascaded Integrated-comb (CIC) Filter (License: IP-CIC)

☐ Intel® FPGA IP Reed-Solomon Encoder/Decoder II (License: IP-RSCODECII)

☐ Intel® FPGA IP Parallel High-Speed Viterbi Decoder (License: IP-VITERBI/HS)

☐ Intel® FPGA IP Serial Low-Speed Viterbi Decoder (License: IP-VITERBI/SS)

☐ ModelSim*-Intel® FPGA Edition Software (License: SW-MODELSIM-AE)

☒ Questa*-Intel® FPGA Edition (License: SW-QUESTA-PLUS)

☐ Intel® FPGA Non-volatile Encryption Key Programming IP (License: SII-NONVOLENCRYPT)

☒ Intel® FPGA AI Suite (License: SW-FPGAISUITE)

* # of Seats

5.2.4.3 Click Next

5.2.5 Add a New Computer

5.2.5.1 Click on +New Computer

Add Host & Generate License

* Generate License (Create a New Computer Or Choose an Existing Computer)

Choose an Existing Computer [View All Computers](#)

Create a New Computer [+New Computer](#)

5.2.5.2 Create Computer

- Computer Name: Enter a unique name
- License Type: Fixed
- Computer Type: NIC ID
- Primary Computer: Enter 12 digit HEX value of the computer NIC ID

Create Computer

* Computer Name ⓘ My computer * License Type ⓘ FIXED Companion Computer ID 1 ⓘ * Primary Admin ⓘ [REDACTED]	* Computer Type ⓘ NIC ID ⓘ How to find your hardware information (NIC/Host/Guard ID)? * Primary Computer ID ⓘ ABCD12345678 Companion Computer ID 2 ⓘ
--	--

Cancel
Save

5.2.5.3 Press Save

5.2.5.4 Select the created computer

5.2.5.5 Click the License check box

5.2.5.6 Press Generate

Save the License file

* Generate License (Create a New Computer Or Choose an Existing Computer)

Choose an Existing Computer ⓘ

[View All Computers](#)

☒

ABCD12345678

My computer

Create a New Computer ⓘ

+New Computer

* ☒ I have read and agree to the terms of use of this license as listed below

This is for 90-Day Evaluation License License includes: •Evaluation version of the Quartus Prime Standard Edition •Evaluation version of the Quartus Prime Pro Edition •FPGA IP Base Suite •DSP Builder for FPGAs •Quartus Prime Separation Feature •Mentor Graphics® AXI® Verification IP Suite – FPGA Edition •Questa®-FPGA Edition – Questa Plus •FPGA AI Suite

☐ Check this box if you don't want Intel to contact you for feedback. Your feedback helps us improve the product.

Back
Generate

5.2.6 Add a Nios V/m IP license

5.2.6.1 Select **Sign up for Evaluation or No-Cost Licenses** again

Intel® FPGA Self-Service Licensing Center

▼ Admins ▼
Sign up for Evaluation or No-Cost Licenses

5.2.6.2 Select the Nios V/m Microcontroller FPGA IP



5.2.6.3 Click **Next**

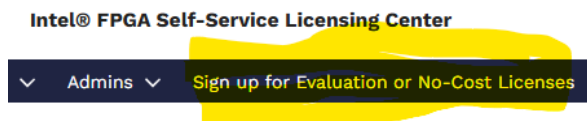
5.2.6.4 Select the **existing computer** that was created in 5.2.5.2

5.2.6.5 Click the License check box

5.2.6.6 Press Generate

5.2.7 Add a MIPI DSI/CSI-2 FPGA IP license

5.2.7.1 Select **Sign up for Evaluation or No-Cost Licenses** again



5.2.7.2 Select the MIPI DSI/CSI-2 FPGA IP



5.2.7.3 Click **Next**

5.2.7.4 Select the **existing computer** that was created in 5.2.5.2

5.2.7.5 Click the License check box

5.2.7.6 Press Generate

* Generate License (Create a New Computer Or Choose an Existing Computer)

Choose an Existing Computer ⓘ
[View All Computers](#)

ABCD12345678
 My computer

Create a New Computer ⓘ
 +New Computer

* ☒ I have read and agree to the terms of use of this license as listed below
 This is for 90-Day Evaluation License License includes: •Evaluation version of the Quartus Prime Standard Edition •Evaluation version of the Quartus Prime Pro Edition •FPGA IP Base Suite •DSP Builder for FPGAs •Quartus Prime Separation Feature •Mentor Graphics* AXI* Verification IP Suite – FPGA Edition •Questa*-FPGA Edition - Questa Plus •FPGA AI Suite

☐ Check this box if you don't want Intel to contact you for feedback. Your feedback helps us improve the product.

Back Generate

5.2.8 Save the License files

A total of five emails and licenses will be sent to the email address on record.

5.2.8.1 Save the **last two** licenses sent. The remainder are **duplicates**. A good location could be **c:\altera_pro**

5.2.9 Merge the two license files into one

5.2.9.1 Open the **two license** files in a text editor (Notepad or Notepad++ for example)

5.2.9.2 **Cut and paste** the contents of the first license file into the second license file.

5.2.9.3 Make a note of the second license **filename** and save it.

5.2.10 Add Environment variables for Quartus and Questa

5.2.10.1 Open the System Properties window by pressing the **Windows key** and "R". This will open the Run window.

5.2.10.2 Type the following Run command: **SystemPropertiesAdvanced** and press **enter**

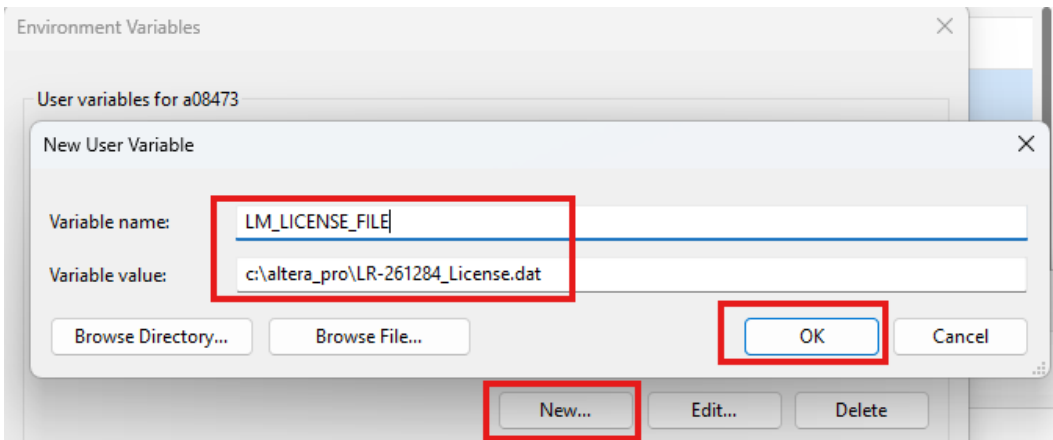
5.2.10.3 Click the Environment Variables **button**

5.2.10.4 Press **New** to create a new User Environment Variable

5.2.10.5 Enter **LM_LICENSE_FILE** for the **Variable name**

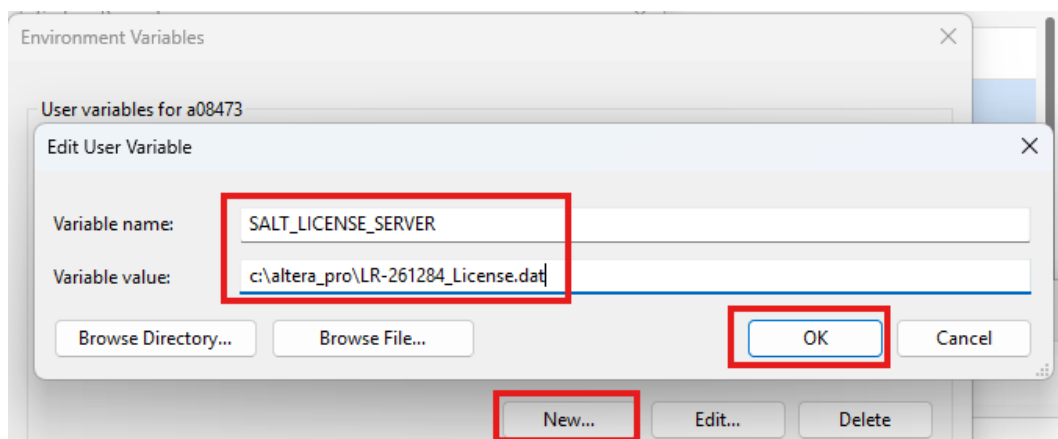
5.2.10.6 Enter the path to the license file for the **Variable value**

5.2.10.7 Press OK



5.2.10.8 Press **New** to add a new environment variable.

5.2.10.9 Repeat the steps above to add the **Questa** environment variable



5.2.10.10 Press OK twice to exit System Properties

5.2.11 Set up the Quartus License

5.2.11.1 Open Quartus

5.2.11.2 Open the Quartus License setup page. Tools → License Setup

5.2.11.3 Navigate to the **Use LM_LICENSE_FILE variable**. Click the **check box**

5.2.11.4 Press OK

License Setup

License file: c:\altera_pro\LR-261284_License.dat

☒ Use LM_LICENSE_FILE variable: c:\altera_pro\LR-261284_License.dat

Current license

Mode: Licensed

Expiration: 20-Jan-2026

Host ID Type: NIC ID

Host ID Value: 32e3a4395f94

Download License

Begin 30-day Grace Period

☒ Wait for floating licenses

Get no-cost licenses

☐ Automatically Refresh License Server Token

Licensed AMPP/IP cores functions:

Vendor	Product	Version	Expiration	Count	Hostid	Vz
Altera (6AF7)	DDR2 High Performance Memory Controller (00BF)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	DDR3 High Performance Memory Controller (00C2)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (C4D5)	DSP Builder (512A)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	FFT/IFFT (0034)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	FIR Compiler II (00D8)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF8)	Intel FPGA AI Suite (018B)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	Intel FPGA AI Suite (018B)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	BCH Codec (00FD)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	Reed Solomon High Speed Codec (00FC)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF8)	10Gb Ethernet IP Reference Design (E003)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	10Gb Ethernet IP Reference Design (E003)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	0114	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	Base CXL Hard IP with spec-defined subprotocol interfaces (AVST) (0186)	2026.01	20-Jan-2026	Fixed	32e3a439	
Altera (6AF7)	CXL IP for Device Type1 with optional built-in cache (0187)	2026.01	20-Jan-2026	Fixed	32e3a439	

5.2.12 Set the environment variable for Questa

5.2.12.1 Create a new environment variable named SALT_LICENSE_SERVER

5.2.12.2 Point to the License file on the computer.

6 Legal Disclaimer

ARROW ELECTRONICS

EVALUATION BOARD LICENSE AGREEMENT

By using this evaluation board or kit (together with all related software, firmware, components, and documentation provided by Arrow, "Evaluation Board"), You ("You") are agreeing to be bound by the terms and conditions of this Evaluation Board License Agreement ("Agreement"). Do not use the Evaluation Board until You have read and agreed to this Agreement. Your use of the Evaluation Board constitutes Your acceptance of this Agreement.

PURPOSE

The purpose of this evaluation board is solely intended for evaluation purposes. Any use of the Board beyond these purposes is on your own risk. Furthermore, according to the applicable law, the offering Arrow entity explicitly does not warrant, guarantee or provide any remedies to you with regard to the board.

LICENSE

Arrow grants You a non-exclusive, limited right to use the enclosed Evaluation Board offering limited features only for Your evaluation and testing purposes in a research and development setting. Usage in a live environment is prohibited. The Evaluation Board shall not be, in any case, directly or indirectly assembled as a part in any production of Yours as it is solely developed to serve evaluation purposes and has no direct function and is not a finished product.

EVALUATION BOARD STATUS

The Evaluation Board offers limited features allowing You only to evaluate and test purposes. The Evaluation Board is not intended for consumer or household use. You are not authorized to use the Evaluation Board in any production system, and it may not be offered for sale or lease, or sold, leased or otherwise distributed for commercial purposes.

OWNERSHIP AND COPYRIGHT

Title to the Evaluation Board remains with Arrow and/or its licensors. This Agreement does not involve any transfer of intellectual property rights ("IPR") for evaluation board. You may not remove any copyright or other proprietary rights notices without prior written authorization from Arrow or its licensors.

RESTRICTIONS AND WARNINGS

Before You handle or use the Evaluation Board, You shall comply with all such warnings and other instructions and employ reasonable safety precautions in using the Evaluation Board. Failure to do so may result in death, personal injury, or property damage.

You shall not use the Evaluation Board in any safety critical or functional safety testing, including but not limited to testing of life supporting, military or nuclear

applications. Arrow expressly disclaims any responsibility for such usage which shall be made at Your sole risk.

WARRANTY

Arrow warrants that it has the right to provide the evaluation board to you. This warranty is provided by Arrow in lieu of all other warranties, written or oral, statutory, express or implied, including any warranty as to merchantability, non-infringement, fitness for any particular purpose, or uninterrupted or error-free operation, all of which are expressly disclaimed. The evaluation board is provided “as is” without any other rights or warranties, directly or indirectly.

You warrant to Arrow that the evaluation board is used only by electronics experts who understand the dangers of handling and using such items, you assume all responsibility and liability for any improper or unsafe handling or use of the evaluation board by you, your employees, affiliates, contractors, and designees.

LIMITATION OF LIABILITIES

In no event shall Arrow be liable to you, whether in contract, tort (including negligence), strict liability, or any other legal theory, for any direct, indirect, special, consequential, incidental, punitive, or exemplary damages with respect to any matters relating to this agreement. In no event shall arrow’s liability arising out of this agreement in the aggregate exceed the amount paid by you under this agreement for the purchase of the evaluation board.

IDENTIFICATION

You shall, at Your expense, defend Arrow and its Affiliates and Licensors against a claim or action brought by a third party for infringement or misappropriation of any patent, copyright, trade secret or other intellectual property right of a third party to the extent resulting from (1) Your combination of the Evaluation Board with any other component, system, software, or firmware, (2) Your modification of the Evaluation Board, or (3) Your use of the Evaluation Board in a manner not permitted under this Agreement. You shall indemnify Arrow and its Affiliates and Licensors against and pay any resulting costs and damages finally awarded against Arrow and its Affiliates and Licensors or agreed to in any settlement, provided that You have sole control of the defense and settlement of the claim or action, and Arrow cooperates in the defense and furnishes all related evidence under its control at Your expense. Arrow will be entitled to participate in the defense of such claim or action and to employ counsel at its own expense.

RECYCLING

The Evaluation Board is not to be disposed as an urban waste. At the end of its life cycle, differentiated waste collection must be followed, as stated in the directive 2002/96/EC. In all the countries belonging to the European Union (EU Dir. 2002/96/EC) and those following differentiated recycling, the Evaluation Board is subject to differentiated recycling at the end of its life cycle, therefore: It is forbidden

to dispose the Evaluation Board as an undifferentiated waste or with other domestic wastes. Consult the local authorities for more information on the proper disposal channels. An incorrect Evaluation Board disposal may cause damage to the environment and is punishable by the law.